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Hi Harshavardhan
192511771HD ▾

QUESTIONS 2 / 5 completed

Insertion Sort ▾

INSERTION SORT

Write a program sort the elements 9, 5, 1, 4, 3 using Insertion Sort.

PROGRAM EDITOR C ▾ Run Save

```
1 #include <stdio.h>
2
3 int main() {
4     int arr[] = {9, 5, 1, 4, 3};
5     int n = 5;
6     int i, key, j;
7
8     for (i = 1; i < n; i++) {
9         key = arr[i];
10        j = i - 1;
11
12        while (j >= 0 && arr[j] > key) {
13            arr[j + 1] = arr[j];
14            j = j - 1;
15        }
16        arr[j + 1] = key;
17    }
18
19    for (i = 0; i < n; i++) {
20        printf("%d ", arr[i]);
21    }
22 }
```

OUTPUT

1 3 4 5 9

INPUT

9 5 1 4 3

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8
9     while (j >= 0 && arr[j] > key) {
10         arr[j + 1] = arr[j];
11         j = j - 1;
12     }
13     arr[j + 1] = key;
14 }
15
16 for (i = 0; i < n; i++) {
17     printf("%d ", arr[i]);
18 }
19
20 return 0;
21 }
```

OUTPUT

1 3 4 5 9

INPUT

9 5 1 4 3